

Cyclic Quartic Root Inequality

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Problem

For all real numbers a, b, c , prove that

$$\sqrt[4]{\frac{a^2}{a^2 - ab + b^2}} + \sqrt[4]{\frac{b^2}{b^2 - bc + c^2}} + \sqrt[4]{\frac{c^2}{c^2 - ca + a^2}} \leq 3.$$

Solution

Let

$$f(x, y) = \sqrt[4]{\frac{x^2}{x^2 - xy + y^2}}.$$

Since

$$x^2 - xy + y^2 = |x|^2 - |x||y| + |y|^2,$$

we have

$$f(x, y) \leq \sqrt[4]{\frac{|x|^2}{|x|^2 - |x||y| + |y|^2}}.$$

Hence it suffices to prove the inequality for nonnegative a, b, c .

If one of the variables is zero, say $c = 0$, then

$$f(a, b) + f(b, 0) + f(0, a) = f(a, b) + 1.$$

Moreover,

$$\frac{a^2}{a^2 - ab + b^2} = \frac{1}{\left(\frac{b}{a} - \frac{1}{2}\right)^2 + \frac{3}{4}} \leq \frac{4}{3},$$

and therefore

$$f(a, b) \leq \left(\frac{4}{3}\right)^{1/4} < 2.$$

Thus

$$f(a, b) + 1 < 3,$$

and the result follows.

We may therefore assume that $a, b, c > 0$.

Observe that

$$x^2 - xy + y^2 = \frac{(x + y)^2 + 3(x - y)^2}{4}.$$

Hence

$$f(x, y) = \sqrt[4]{\frac{4x^2}{(x + y)^2 + 3(x - y)^2}} \leq \sqrt[4]{\frac{4x^2}{(x + y)^2}} = \sqrt{\frac{2x}{x + y}}.$$

Applying this estimate cyclically gives

$$\sum f \leq \sqrt{\frac{2a}{a + b}} + \sqrt{\frac{2b}{b + c}} + \sqrt{\frac{2c}{a + c}}.$$

By the Cauchy–Schwarz inequality,

$$\left(\sqrt{\frac{2a}{a + b}} + \sqrt{\frac{2b}{b + c}} + \sqrt{\frac{2c}{a + c}} \right)^2$$

$$\leq (a + b + c) \left(\frac{2a}{(a + b)(a + c)} + \frac{2b}{(b + c)(a + b)} + \frac{2c}{(a + c)(b + c)} \right).$$

A routine simplification yields

$$= \frac{2(a + b + c)(ab + bc + ca)}{(a + b)(b + c)(c + a)}.$$

Using

$$(a + b)(b + c)(c + a) = (a + b + c)(ab + bc + ca) - abc,$$

we obtain

$$= 2 + \frac{2abc}{(a + b)(b + c)(c + a)}.$$

Finally, by AM–GM,

$$(a + b)(b + c)(c + a) \geq (2\sqrt{ab})(2\sqrt{bc})(2\sqrt{ca}) = 8abc.$$

Therefore

$$\frac{2abc}{(a + b)(b + c)(c + a)} \leq \frac{1}{4},$$

and so

$$\sum f \leq 2\sqrt{2 + \frac{1}{4}} = 2\sqrt{\frac{9}{4}} = 3.$$

Hence

$$\boxed{\sqrt[4]{\frac{a^2}{a^2 - ab + b^2}} + \sqrt[4]{\frac{b^2}{b^2 - bc + c^2}} + \sqrt[4]{\frac{c^2}{c^2 - ca + a^2}} \leq 3}.$$